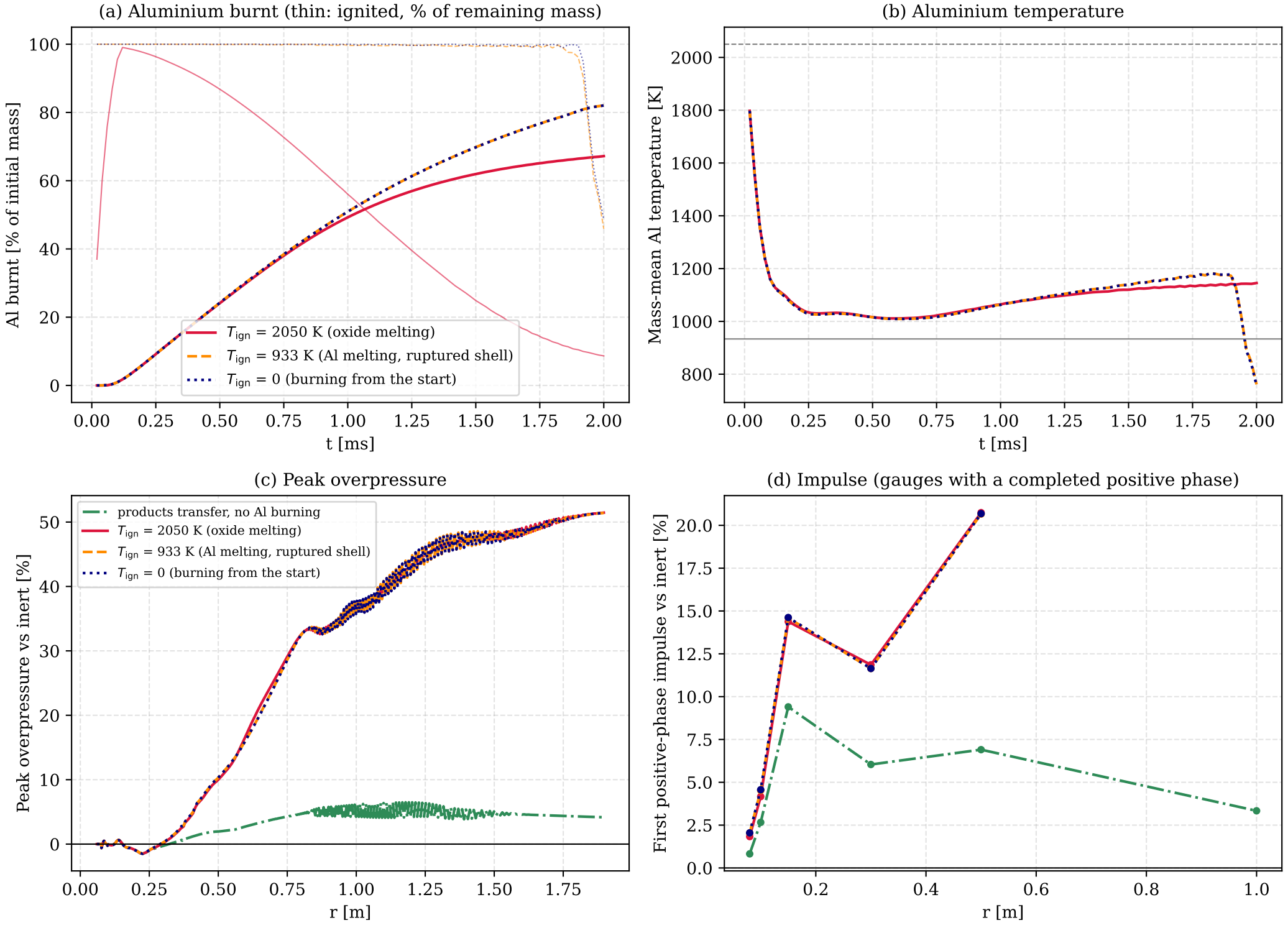
(a) Aluminium burnt (thin: ignited, % of remaining mass)
Al burnt [% of initial mass]
t [ms]
T_{ign} = 2050 K (oxide melting)
T_{ign} = 933 K (Al melting, ruptured shell)
T_{ign} = 0 (burning from the start)
(b) Aluminium temperature
Mass-mean Al temperature [K]
t [ms]
(c) Peak overpressure
Peak overpressure vs inert [%]
r [m]
products transfer, no Al burning
T_{ign} = 2050 K (oxide melting)
T_{ign} = 933 K (Al melting, ruptured shell)
T_{ign} = 0 (burning from the start)
(d) Impulse (gauges with a completed positive phase)
First positive-phase impulse vs inert [%]
r [m]